point signal, $x[n]$. Taking the DFT provides the frequency spectrum contained in $ReX[k]$ and $ImX[k]$, where k runs from 0 to $N/2$. However, these are not the amplitudes needed to reconstruct the time domain waveform; samples $ReX[0]$ and $ReX[N/2]$ must first be divided by two. (See Eq. 8-3 to refresh your memory). This is easily carried out in computer programs, but inconvenient to deal with in equations.

The complex Fourier transform is an elegant solution to these problems. It is natural for complex numbers and negative frequencies to go hand-in-hand. Let's see how it works.

Mathematical Equivalence

Our first step is to show how sine and cosine waves can be written in an *equation* with complex numbers. The key to this is Euler's relation, presented in the last chapter:

EQUATION 31-2
Euler's relation.

$$e^{jx} = \cos(x) + j\sin(x)$$

At first glance, this doesn't appear to be much help; one complex expression is equal to another complex expression. Nevertheless, a little algebra can rearrange the relation into two other forms:

EQUATION 31-3
Euler's relation for sine & cosine.

$$\cos(x) = \frac{e^{jx} + e^{-jx}}{2} \qquad \sin(x) = \frac{e^{jx} - e^{-jx}}{2j}$$

This result is extremely important, we have developed a way of writing *equations* between complex numbers and ordinary sinusoids. Although Eq. 31-3 is the standard form of the identity, it will be more useful for this discussion if we change a few terms around:

EQUATION 31-4
Sinusoids as complex numbers. Using complex numbers, cosine and sine waves can be written as the sum of a positive and a negative frequency.

$$\cos(\omega t) = \frac{1}{2}e^{j(-\omega)t} + \frac{1}{2}e^{j\omega t}$$

$$\sin(\omega t) = \frac{1}{2}je^{j(-\omega)t} - \frac{1}{2}je^{j\omega t}$$

Each expression is the sum of two exponentials: one containing a *positive* frequency (ω), and the other containing a *negative* frequency (-ω). In other words, when sine and cosine waves are written as complex numbers, the